

Exercise: Exploring Your Home LAN Network

Objective

Discover and analyze the Local Area Network (LAN) structure in your home by identifying network devices, topology, and addressing schemes. This hands-on exercise will help you understand how real-world networks are organized.

Learning Outcomes

By completing this exercise, you will be able to:

- Identify network devices and their roles in your home network
- Determine the network topology being used
- Find and interpret physical (MAC) and logical (IP) addresses
- Understand how devices communicate within a LAN
- Create a network diagram based on real infrastructure

Materials Needed

- Your home computer/laptop (Windows, Mac, or Linux)
- Access to your home router/modem
- Smartphone or tablet (optional)

Part 1: Physical Device Discovery

Tasks:

1. Locate your networking hardware:

- Find your modem (connects to ISP) and router (may be combined in one device)
- Note the model and manufacturer
- Take a photo or sketch the device, showing the ports and connections

2. Count connected devices:

- List all devices connected to your network (computers, phones, smart TVs, IoT devices, printers, etc.)
- Identify how each device connects: wired (Ethernet cable) or wireless (Wi-Fi)

3. Examine physical connections:

- Look at the back of your router - count the Ethernet ports

- If possible, trace where Ethernet cables lead
- Observe any additional networking equipment (switches, access points, repeaters)

Part 2: Network Configuration Analysis

1. Open Command Prompt (search "cmd")
2. Type ipconfig /all and press Enter
3. Screenshot or write down, explain what are these and their format:
 - Your computer's IP address:
 - Subnet mask
 - Default gateway (router's IP address)
 - Physical address (MAC address)

Part 3: Network Topology Identification

Tasks:

1. Based on your observations, identify the topology:
 - How are devices connected to the router?
 - Is there a central device that all others connect through?
 - Are devices connected in a line, or do they each have independent connections?
2. Determine if you have:
 - Bus topology
 - Pure star topology
 - Extended star (star with additional switches)
 - Hybrid topology (combination of types)

Part 4: Create a Network Diagram

Tasks: Draw a complete network diagram that includes:

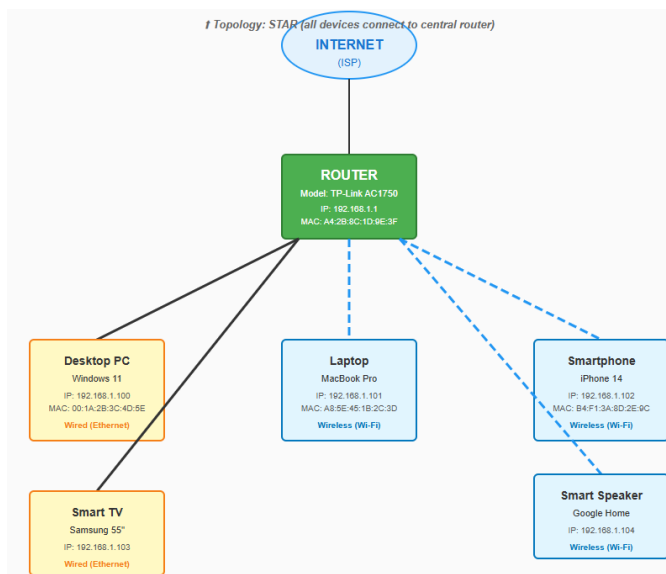
- Your ISP connection (external)
- Modem/Router (label with model and IP address)
- All connected devices with:
 - Device name/type

- Connection type (wired/wireless)
- IP address
- MAC address (for at least 2-3 devices)
- Any additional networking equipment (switches, access points)
- Use appropriate symbols or shapes to differentiate device types

Diagram tips:

- Place the router/switch at the center
- Draw lines showing connections (solid for wired, dashed for wireless)
- Label everything clearly
- Use colors or different line styles to make it more readable

Example diagram:



Part 5: Analysis Questions

Answer the following questions based on your exploration:

1. **Topology:** What network topology does your home network use? Justify your answer.
2. **Addressing:**
 - What is the IP address range used in your network?
 - Are the IP addresses private or public? How can you tell?
 - How do you think devices receive their IP addresses?
3. **Central device:**

- What is the role of your router in the network?
- What would happen if the router stopped working?

4. Scalability:

- How many devices can your network support?
- What would you need to add if you wanted to connect more devices than available ports?

5. Comparison:

- How does your home topology compare to bus topology?
- What are the advantages of your current setup?

Deliverables

Submit the document include the following:

1. Network diagram with all required labels
2. List of all devices with their IP and MAC addresses (at least your own device)
3. Screenshots of your ipconfig or ifconfig output
4. Photos of your router/networking equipment (optional)
5. Written answers to all analysis questions (1-2 paragraphs each)

Your submission should be written in Microsoft Word in .docx format and be named with <StudentID>.docx where <StudentID> is your Student ID (for example 25123456.docx).